Problem #4 Free Cell Game Design Report

Overview

freecellPlayer(clock, source, dest, win) stores one free cell game in registers and performs one move per rising clock edge. It starts with game 8321, which is loaded with an initial block that ignores illegal moves and declares a win when all 52 cards are home. It is named freecellPlayer to match the provided testbench testFreecell.v.

State Representation

Each card is a 6-bit code {suit, rank}: suit H = 00, D = 01, S = 10, C = 11 and rank A = 1 to K = 13. The code 0 means "no card", so every slot is its own empty flag, and bit 5 is the color bit (0 = red, 1 = black).

Storage	Declaration
Tableau columns	reg [5:0] col [0:7][0:18]
Column lengths	reg [4:0] len [0:7]
Free cells	reg [5:0] fc [0:3]
Home cells (top rank per suit)	reg [3:0] home [0:3]

Each column is a stack with a movable card at col[i][len[i]-1]. 19 slots is the most a column can hold since you account for 7 dealt plus a run of 12. The home cell stores only the rank of its top card since the cards beneath it are always A up to that rank.

Move Processing

Each move is handled inside the always @(posedge clock) block where it first finds the card being moved, either from the head of the source column or from the free cell. An empty source or an illegal source code gives a card value of 0 and is markedillegal. Then, it checks whether the destination can accept the card. A column accepts it if the column is empty or if its head card is the opposite color and one rank higher. A free cell accepts it only if the cell is empty and a home cell accepts it only if home[suit] + 1 == rank. Finally, if the move is legal, the card is removed from the source and placed at the destination, but If not legal, the state stays the same. The win output goes to 1 once all four home cells equal 13.

Testing and Efficiency

The design passes the provided testFreecell.v without changes. The testbench plays 115 legal moves of game 8321 with 6 illegal moves. The design rejects those 6 moves, and win rises on the final move at t = 1205. The design is simple and efficient because finding the movable card takes one array read, and each legality check is only a few comparisons. Storing each card’s location instead would use less storage, but every move would then require a search through all 52 cards. The design also saves space by storing only the top rank in each home cell instead of every card and by using 0 as the empty flag.